SQL Detective Game
Unity Seminar – Project Proposal

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Introduction

Our project is a 3D educational game designed to teach SQL query-building through story-driven gameplay. The game immerses players in a dynamic virtual environment where they take on the role of an investigator navigating a data-rich world. By solving SQL-based puzzles, players will learn how to construct queries, analyse tables, and retrieve meaningful data to progress in the game.

Purpose Of The Project

In today's world, where the volume of data grows exponentially alongside our reliance on it, the need to effectively manage, retrieve, and analyse information has become a critical challenge. This creates an essential goal: enabling the effective navigation, retrieval, and analysis of data to unlock its full potential, but accessing it requires foundational skills.

Many people interested in entering the world of data lack experience and need an introduction to SQL and data management. Furthermore, teaching children to navigate and retrieve information from large datasets from an early age is essential for their personal development and crucial for society.

Our project addresses these gaps by providing an immersive, story-driven educational experience that teaches SQL query-building through gameplay. By simulating a rich data environment, players will learn to navigate databases, construct queries, and extract insights in a hands-on manner.
In addition, It has already been proven that teaching a subject in the form of a fun activity such of a game has a greater impact on remembering the materials and understanding them, which is why this game has an edge over other non-interactive SQL teaching materials.

Target Audience

Our target audience includes a diverse group of people new to SQL who need a platform that immerses them in an authentic environment, allowing them to feel more connected to the data they work with, as well for children, highlighting the importance of introducing data literacy and SQL skills at an early age.
Additionally, the game appeals to casual gamers who enjoy story-driven thinking games, blending entertainment with educational value.

Market Analysis

The number of SQL-based educational games available on the market is limited.
our game stands out with key differences that set it apart:

1. 3D Environment: Most SQL games are in 2D or have a simple interface, but our game puts players in a fully immersive 3D environment. It feels more realistic and gives players a better sense of actually working with data in a real-world scenario.

2. Story-Driven Scenarios: The game offers dynamic clues and missions that change based on the player's progress. Each session provides fresh challenges and integrates SQL tasks naturally into the storyline, keeping players engaged while learning.

3. visualization data - Players can analyse their data through graphs, charts, and other visuals, helping them identify trends or patterns. This feature makes it easier to understand the data and bridges the gap between raw query outputs and meaningful insights, like BI tools.

Main Features

1. Introduction to the game

At the start, Main Menu will be represented and include:

1. Start Game
   1.1. Start
   1.2. Choose difficulty
2. Continue saved game
3. Tutorials
4. View Leaderboard
5. Exit

At the start, the player will be introduced to the game's plot, the purpose of the game, objectives, and mechanics.
The databases and their attributes will be explained, along with a relational diagram to help players understand the schema.
The databases will also be displayed later in the game when the player wants to perform a retrieval.

Frameworks/Tools: Unity and UI

2. Movement and Environment

The game will feature a Third-Person View.
Players will navigate a detailed 3D environment, and interact with objects and clues to progress, creating a realistic learning and gaming experience

Frameworks/Tools: Unity's 3D, Unity Animation system, assets for Unity's Assets Store

3. Game's plot and Databases

The game places the player in the role of a detective solving crime mystery through interacting with objects, collecting clues, and using SQL-driven puzzles to interrogate databases and progress through the story, and each session revolves around investigating a unique case.

The databases in the game collectively form a schema presented to the player as a visual map.
By becoming familiar with the schema and databases, the player will build confidence in understanding the data structure, which allows him to focus more on developing SQL skills as the queries become more complex.
As the game advances, the player will be exposed to more parts of the schema, and for higher difficulty levels, the player may encounter entirely new schemas, requiring him to adapt and test orientation, problem-solving, and cross-referencing
skills in unfamiliar "worlds".

4. SQL Query Mechanics

The game progresses through solving questions using SQL data retrieval.
The SQL mechanics are supported by an intuitive interface designed to teach and challenge players as they build and execute queries. This interface includes the following components:
- **Query Panel:** An empty panel where the player constructs the query, accompanied by an "EXECUTE" button to run the retrieval operation.
- **Reserved Words Panel:** A panel displaying key SQL reserved words such as SELECT, FROM, WHERE, JOIN, aggregate functions, and more.
- **Tables List:** A list of tables available to the player for retrieval

Players can construct their queries in two ways:
1. Dragging: Players drag reserved words and tables to the query panel. After selecting SELECT and FROM, players can choose fields from the selected table to display, and add conditions like WHERE clauses or aggregate functions.
2. Manual Entry: Players can write the SQL query directly using the keyboard, providing a more advanced challenge for experienced users.

In addition to query-building, players are provided with a "suspects" table, which serves as a dynamic tool for tracking and managing their investigation. Players can:
1. Add Records: Use INSERT INTO to add individuals they consider suspicious.
2. Delete Records: Use DELETE to remove individuals they rule out as suspects.
3. Retrieve Records: Query the suspects table to guess the final answer - who they believe committed the crime. Incorrect guesses result in losing a life and deducting points based on the scoring system, adding stakes to the player's choices.

Frameworks/Tools: Kaggle, SQLite / SQL Server

5. Visualization and BI tools

Visualization complements data retrieval by turning raw data into actionable insights, helping users identify patterns, trends, and see their data “comes to life”.
In the game - after executing a query, the player can display the results visually. For example:
1. Selecting Users by city could generate a bar chart showing user counts per city.
2. A heat map displaying the number of people passing a crime scene at the same day in a week and at the same hour in a day can lead to new suspects or investigative directions.

Frameworks/Tools: Unity UI System / Graph and Chart Asset Pack

6. Scoring and Lives

The scoring and life system adds challenge and motivation to the game by rewarding players for efficiency and penalizing mistakes. Scores are calculated based on the time taken to solve each SQL query, the number of attempts made, and the complexity of the query.
Players also have a limited number of lives, which are lost when making incorrect guesses about the final suspect.

7. Cloud

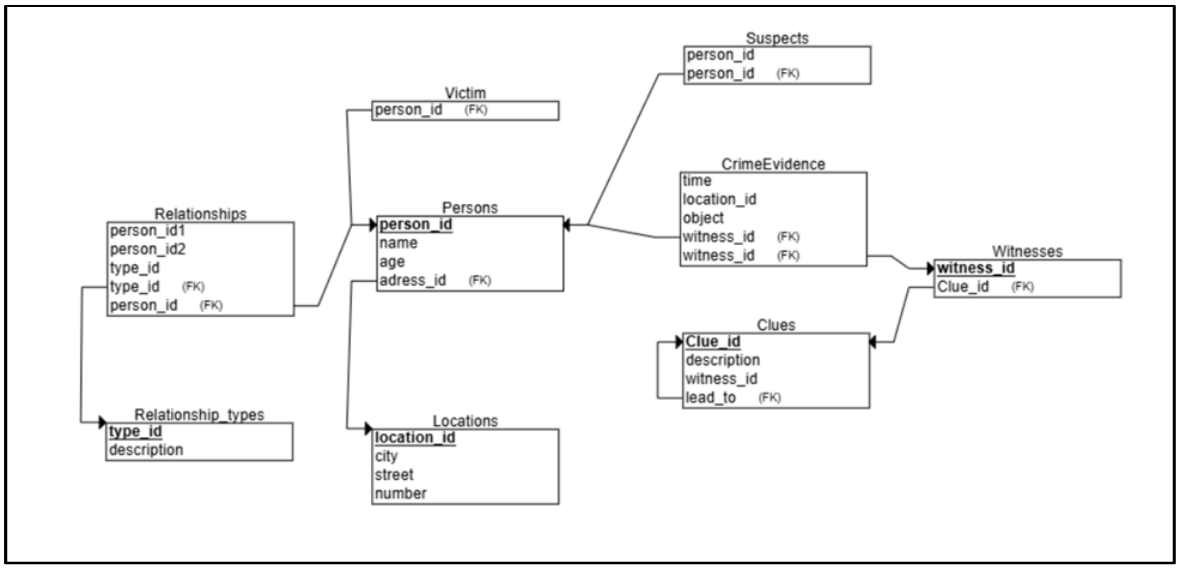
The game uses cloud integration to make sure progress is never lost. Player's progress, including completed levels, scores, and achievements, are saved in the cloud, so players can continue from where they left off on any device.
The cloud also stores dynamic scenarios and leaderboard data, encouraging competition and replayability.

Frameworks/Tools: Realtime databases for cloud saving, Unity Networking, JSON.

Sketches:

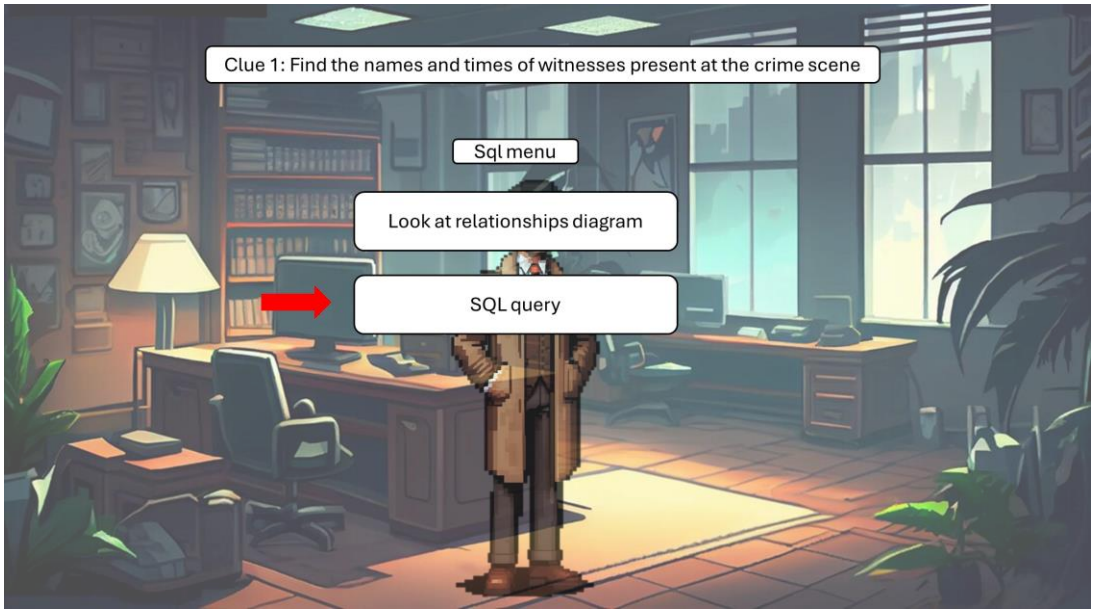
Home page:

An example of the databases and schema shown to the player:

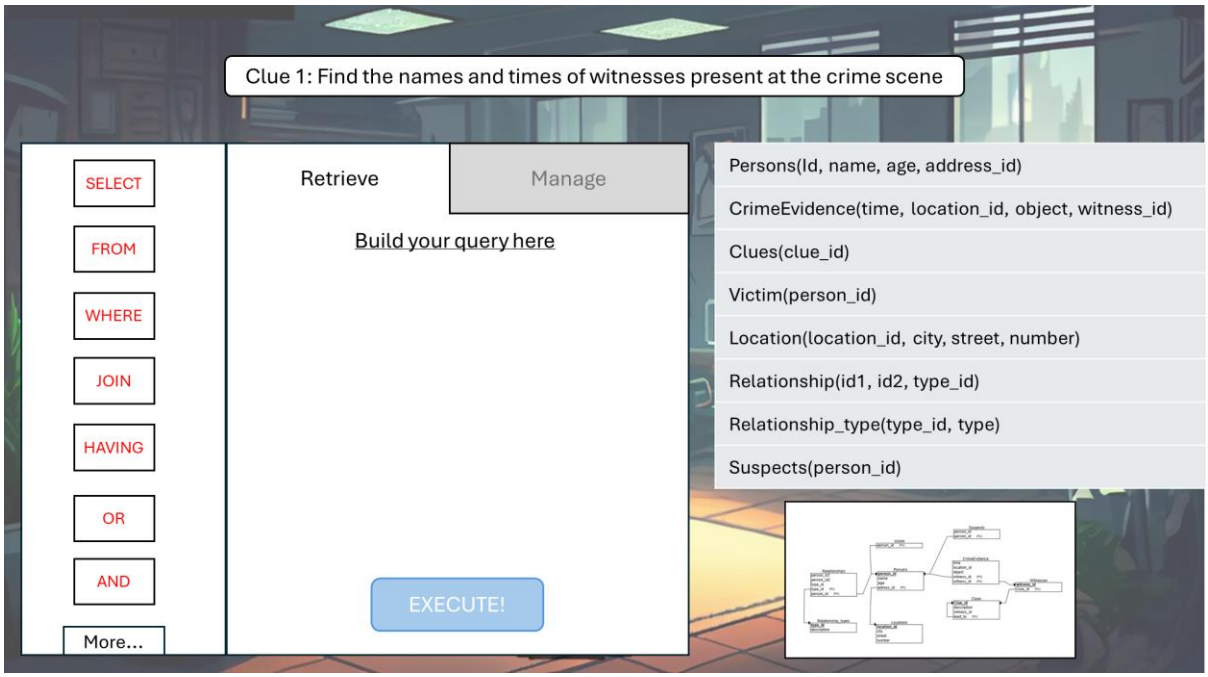


Game screen: the player moves in 3D environment and interacts with objects and UI components

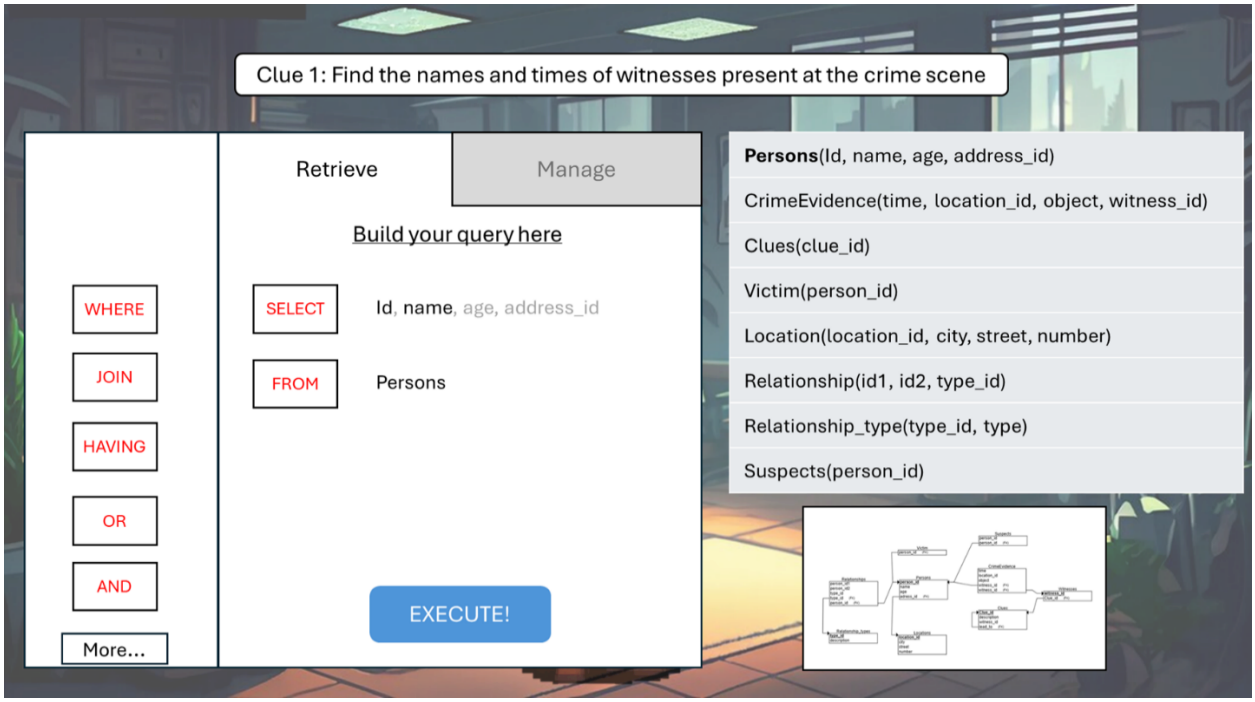
SQL menu:



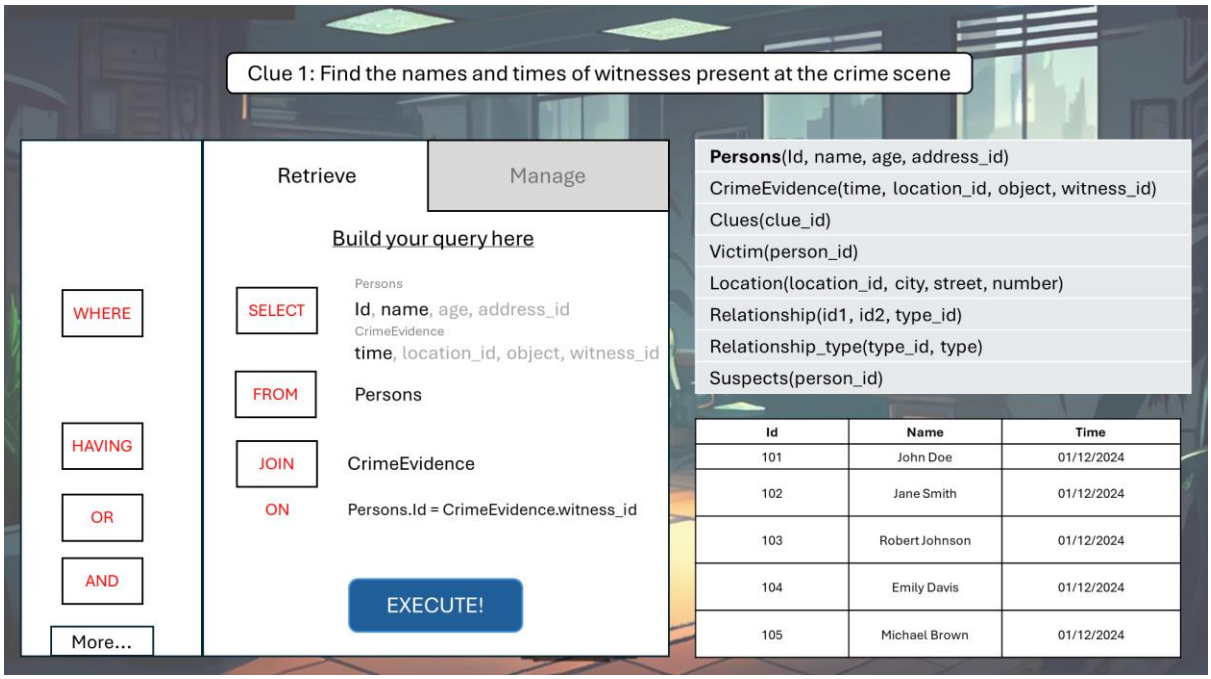
SQL query panel: the player chooses to retrieve or manage Suspect's table



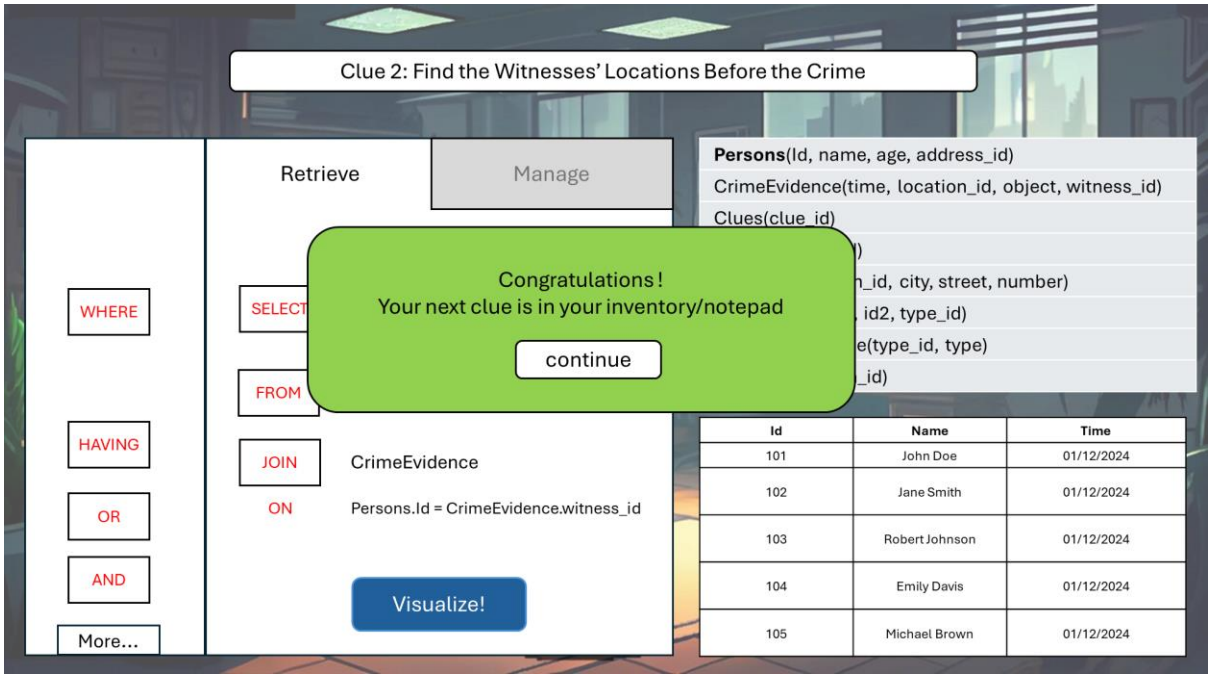
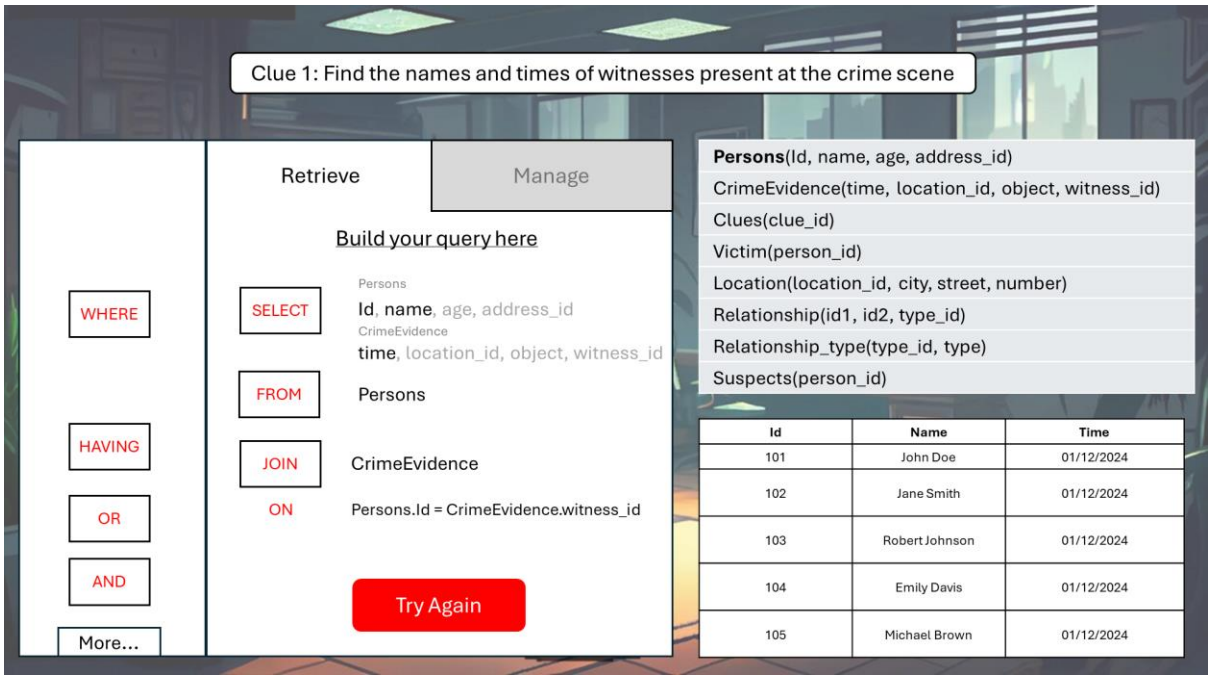
Building query by dragging Reserved Words



Executing the query and output is shown



Option one: output is correct



Option two: output is incorrect